

National, regional, and global estimates of low birthweight in 2020, with trends from 2000: a systematic analysis



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Summary

Background Low birthweight (LBW; <2500 g) is an important predictor of health outcomes throughout the life course. We aimed to update country, regional, and global estimates of LBW prevalence for 2020, with trends from 2000, to assess progress towards global targets to reduce LBW by 30% by 2030.

Methods For this systematic analysis, we searched population-based, nationally representative data on LBW from Jan 1, 2000, to Dec 31, 2020. Using 2042 administrative and survey datapoints from 158 countries and areas, we developed a Bayesian hierarchical regression model incorporating country-specific intercepts, time-varying covariates, non-linear time trends, and bias adjustments based on data quality. We also provided novel estimates by birthweight subgroups.

Findings An estimated 19·8 million (95% credible interval 18·4–21·7 million) or 14·7% (13·7–16·1) of liveborn newborns were LBW worldwide in 2020, compared with 22·1 million (20·7–23·9 million) and 16·6% (15·5–17·9) in 2000—an absolute reduction of 1·9 percentage points between 2000 and 2020. Using 2012 as the baseline, as this is when the Global Nutrition Target began, the estimated average annual rate of reduction from 2012 to 2020 was 0·3% worldwide, 0·85% in southern Asia, and 0·59% in sub-Saharan Africa. Nearly three-quarters of LBW births in 2020 occurred in these two regions: of 19 833 900 estimated LBW births worldwide, 8 817 000 (44·5%) were in southern Asia and 5 381 300 (27·1%) were in sub-Saharan Africa. Of 945 300 estimated LBW births in northern America, Australia and New Zealand, central Asia, and Europe, approximately 35·0% (323 700) weighed less than 2000 g; 5·8% (95% CI 5·2–6·4; 54 800 [95% CI 49 400–60 800]) weighed less than 1000 g, 9·0% (8·7–9·4; 85 400 [82 000–88 900]) weighed between 1000 g and 1499 g, and 19·4% (19·0–19·8; 183 500 [180 000–187 000]) weighed between 1500 g and 1999 g.

Interpretation Insufficient progress has occurred over the past two decades to meet the Global Nutrition Target of a 30% reduction in LBW between 2012 and 2030. Accelerating progress requires investments throughout the lifecycle focused on primary prevention, especially for adolescent girls and women living in the most affected countries. With increasing numbers of births in facilities and advancing electronic information systems, improvements in the quality and availability of administrative LBW data are also achievable.

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Introduction

Low birthweight (LBW) refers to birthweight of less than 2500 g and can occur as a result of preterm birth (before 37 weeks of gestation), intrauterine growth restriction, or both.¹ LBW is associated with early risks, notably of neonatal mortality, and long-term adverse outcomes such as childhood stunting, wasting, neurodevelopmental impairment, and a higher risk of chronic diseases such as type 2 diabetes, hypertension, and cardiovascular disease.^{2–9}

Accelerating efforts to reduce LBW are essential if the associated Global Nutrition Target is to be achieved. This target originally called for a 30% reduction in LBW prevalence—from 15% to 10·5%—between 2012 and 2025. In 2019, WHO and UNICEF called for the target timeline to be extended to 2030, as endorsed by Member States at the World Health Assembly.^{10,11} Achieving this LBW target would also contribute to other Sustainable Development Goals (SDGs), especially those relating to neonatal and child survival, and SDG2 to eliminate

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Research in context

Evidence before this study

In 2019, UNICEF and WHO—in collaboration with the London School of Hygiene & Tropical Medicine, London, UK—published estimates of low birthweight (LBW) for 2000–15, based on systematic data searches and analysis of administrative and household survey data from 2000 using regression modelling. 1447 country-years of birthweight data from 148 countries were used as input data for the model. 20·5 million (95% CI 17·4–24·0 million) LBW livebirths were estimated worldwide in 2015, and these estimates were an advance on survey adjustment methods and estimates published in 2004.

Added value of this study

In this analysis, we expanded the previous LBW database of household surveys and administrative data and applied a Bayesian approach to estimate LBW for all 194 WHO Member States and the occupied Palestinian territory, including East Jerusalem (referred to as countries and areas) from 2000 to 2020. Compared with the previous estimates, ten additional countries contributed LBW input data, totalling 158 countries (81% of 195 countries and areas now covering 75% of global livebirths with LBW data). We systematically selected the covariates using a priori criteria in a four-step process, based on a conceptual framework, data quality assessment of time series, and cluster and correlation analyses. The hierarchical Bayesian framework incorporated LBW and covariate data, hierarchical random country intercepts, and non-linear time trends. We also accounted for data quality of the administrative sources using weighting and bias shifts. All countries and data sources were accounted for in one model. Incorporating these considerations enabled a more flexible model, which resulted in LBW trends

that are comparable between countries or regions and over time. These trends therefore enable meaningful monitoring towards targets. For the first time to our knowledge, we estimated the distribution by LBW subgroups for some regions in 2020 (for regional groupings, see appendix pp 36–37). For northern America, Australia and New Zealand, central Asia, and Europe, we estimated that 65·6% (619 800 of 945 300) of LBW newborns weighed between 2000 g and less than 2500 g, and only 5·8% (54 800 of 945 300) weighed less than 1000 g, although these newborns are the most preterm and the most vulnerable.

Implications of all the available evidence

Over the past two decades, our estimates suggest a minimal reduction in LBW prevalence. The Global Nutrition Target called for a 30% reduction of LBW by 2025—from 15% in 2012 to 10·5% in 2025—which has now been extended to 2030. The 2012–20 average annual rate of reduction is 0·30% and would need to increase 11-fold, to 3·31%, to reach the 2030 target. To increase progress, action is needed to address the two underlying causes of LBW—preterm birth and intrauterine growth restriction—requiring investments particularly to improve women's nutrition, health, and health care. Although the quality and coverage of LBW data have improved, missed opportunities for further improvement remain, especially in routine health information systems, and it is also crucial to better use these data to inform change at national and subnational levels. The LBW subgroups are helpful for individual risk identification and population-level programme planning.

malnutrition.¹² Monitoring progress is essential to ensure targeted timely action.

Despite more than three-quarters of births worldwide occurring in facilities,¹³ routine birthweight data from low-income and lower-middle-income countries have data gaps, including inaccurate capture (eg, with poor-quality scales) and recording and inadequate collation of data within information systems.^{14,15} Birth certificates often include birthweight, and now about 65% of births globally are registered.¹⁶ Therefore, in addition to the collation of data from all relevant sources, data quality assessments and estimation approaches remain necessary to generate comparable national estimates of LBW. The most recent estimates¹⁷ suggested that 20·5 million (95% CI 17·4–24·0 million) neonates were born with LBW in 2015. 91% of these LBW births occurred in low-income and middle-income countries, with almost half (48%) in southern Asia and around a quarter (24%) in sub-Saharan Africa.¹⁷

This Article aims to update estimates of annual LBW prevalence at the country, regional, and global level from 2000 to 2020, using an expanded database including ten

additional countries, based on a new Bayesian approach that enables the integration of information across countries and timepoints. Additionally, an analysis of LBW subgroups was conducted to inform interventions aimed at prevention of LBW as well as clinical decisions and improved care for vulnerable newborns. Recommendations for improved data collection are re-emphasised to help to close data gaps for this important health outcome.

Methods

Overview

We did a systematic search for birthweight data from national administrative sources and household surveys to generate a database of LBW data (figure 1). We estimated LBW prevalence from 2000 to 2020 for all 194 WHO Member States and the occupied Palestinian territory, including East Jerusalem (subsequently referred to as countries and areas)¹⁸ using a hierarchical Bayesian framework. The model included a priori selected covariates and accounted for differences in data quality by adjusting survey input data before modelling and by using weighting and bias shifts within the model. We report national-level

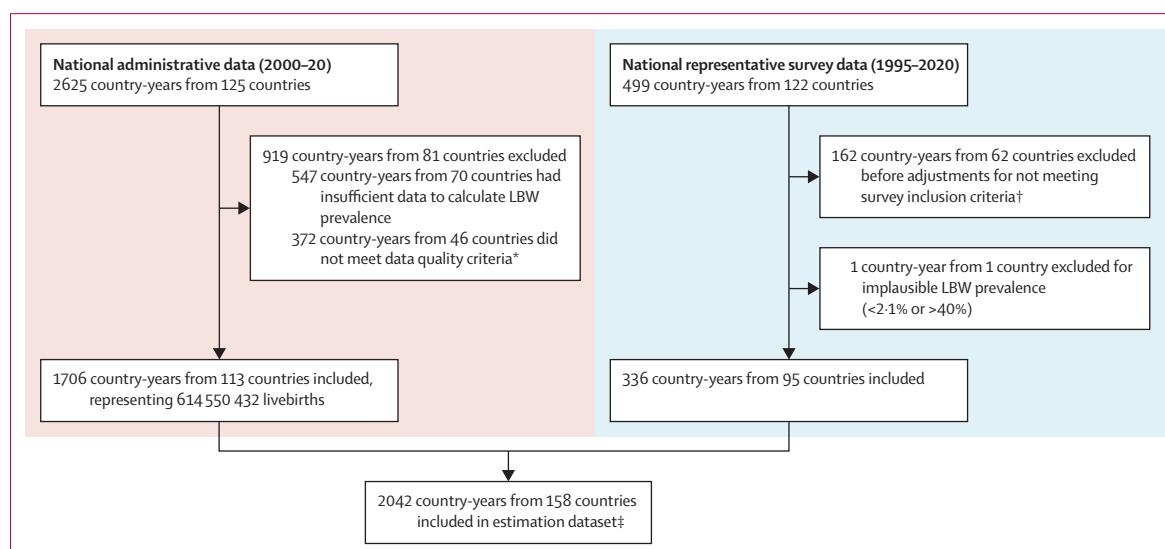


Figure 1: Administrative and survey data inputs for global, regional, and national estimates of LBW for 2000–20

See appendix pp 6–8. LBW=low birthweight. WPP=World Population Prospects. *6 country-years from three countries were excluded owing to an implausible LBW prevalence (<2.1% or >40%); 369 country-years from 45 countries were excluded as they included <80% of WPP-estimated births; 3 country-years from two countries were excluded as they had no information on the number of births in the dataset and were from a context with <80% facility births; some data met more than one exclusion criterion. †Reasons for exclusion were no size at birth and <95% of newborns weighed, percentage weighed did not meet threshold for inclusion, other data quality criteria did not meet threshold for inclusion, model failed, small sample size, and implausible prevalence. ‡Administrative data only: 1127 country-years from 63 countries; survey data only: 151 country-years from 45 countries; administrative and survey data: 762 country-years from 50 countries.

estimates for countries and areas with national-level data meeting the inclusion criteria. We present our methods and results in accordance with the GATHER checklist¹⁹ (appendix pp 3–4). Further details on methods are available in the study protocol²⁰ and the appendix (pp 6–34).

Input data

Data inputs from two main sources were identified and assessed: administrative data from national systems, including civil registration and vital statistics (CRVS) systems, national health management information systems, and birth registries; and nationally representative household surveys, including Demographic Health Surveys, Multiple Indicator Cluster Surveys, and national nutrition surveys for which required variables were available.

Search strategy

For each country meeting a threshold of at least 80% of births in health facilities, we conducted a systematic search of publications and datasets available in the public domain from the relevant Ministry of Health and National Statistical Office.^{21,22} To identify birthweight data for any years between 2000 and 2020, we searched data sources included in the 2015 estimates in addition to conducting a web-based search.¹⁷ For countries without data in the previous estimates, personal communication with government counterparts—facilitated by UNICEF and WHO country offices—took place between September, 2021 and December, 2022.

We searched the UNICEF Nutrition Data Source Catalogue for nationally representative surveys from low-income and middle-income countries (as defined using World Bank classification) from 1998 onwards for which anonymised, individual-level datasets for required variables were available.²⁰

See Online for appendix

Data extraction and processing

Administrative data were abstracted in duplicate by two independent abstractors (two of BK, JC, GG-D, and JK) into a standard database, using a common abstraction guide. Any disagreements between the two abstractors were resolved by a third person (JK or JC). For household surveys, we reanalysed the microdata and produced data quality indicators and LBW prevalence adjusted for missing birthweights and data heaping (appendix pp 6–8) using methods developed for the previous estimates.¹⁷ Analyses were done using Stata version 17.

Exclusion criteria

We excluded administrative data for country-years for which birthweight data were reported for less than 80% of estimated livebirths (according to the UN Population Division's World Population Prospects [WPP] 2022²³). In addition, survey data in which less than 30% of the births in the dataset had a recorded valid birthweight (defined as 250–5500 g), or in which there was indication of severe heaping or implausible birthweight distribution, were also excluded. Data sources with an estimated LBW prevalence of less than 2.1% or greater

than 40% in a given year were considered implausible and excluded, on the basis of the lowest reported prevalence in healthy, low-risk women in the INTERGROWTH-21st project and the highest population-based LBW prevalence recorded in the literature (appendix pp 8–9).^{17,20,24}

UNICEF and WHO country consultation

Country consultations were conducted between Sept 29 and Nov 15, 2022, with National Focal Points nominated by their governments. During the consultation period, focal points had the opportunity to provide new data and data meeting the inclusion criteria were included in the estimates. They also had opportunities to review modelling methods plus preliminary LBW estimates from 2000 to 2020 for their country. Following these consultations, we changed 374 country-years of data from 18 countries and areas that were already in the dataset and added 26 new country-years of data from 11 countries and areas. Inclusion of these new data resulted in changes in LBW prevalence of 3.0–6.1 percentage points for 11 of the smallest of these 18 countries. All other changes were of less than two percentage points. The final dataset comprised 2042 country-years.

Data quality assessment and categorisation

We identified several potential sources of bias in the LBW data and used these a priori²⁰ (appendix pp 11–12) to generate data quality categories for the administrative data (appendix pp 19–20). The potential sources of bias included were recorded birthweight coverage and facility birth rate; data source type (CRVS, medical birth registry, District Health Information Software 2 [DHIS2], or other hospital system); denominator type (livebirths with birthweight, livebirths or total births, or reported LBW prevalence if no data available to calculate); and information on potential omissions of births around the threshold of viability (proportion of LBW newborns <1000 g or <1500 g). Data were categorised by country-year to account for changes over time in data systems (appendix pp 19–20). As well as accounting for the different sources and biases, this categorisation allowed us to adjust the administrative data for heaping by approximating adjustment from the surveys in the same grouping.

Covariate selection

Four steps were applied for covariate selection (appendix pp 13–18): the identification of plausible covariates using a conceptual framework adapted from published frameworks;^{25–28} a search for covariate time series data; data quality assessment of all potential covariates time series, and completing time series using linear interpolation and constant extrapolation; and statistical analysis to identify the smallest set of covariates, including cluster analysis based on correlations between

covariates. The final covariates included in the model were gross national income per person at purchasing power parity (constant 2017 international \$), prevalence of underweight (defined as BMI <18.5 kg/m²) among female adults (aged 15–49 years), adult (aged ≥15 years) female literacy rate, modern contraception prevalence rate, and the percentage of urban population.

Estimation of LBW by year and country

Estimates of LBW prevalence at the national level were predicted from a Bayesian hierarchical regression model (appendix pp 21–25). The model is fit on the logit (log-odds) scale to ensure that proportions are bounded between 0 and 1, and then back-transformed and multiplied by 100 to obtain estimates.

Hierarchical random country-specific intercepts (countries within regions within the global set) accounted for the correlation within and between the regions. The six SDG regions²⁹ were adapted for modelling for this analysis, enabling a few countries to be included in an alternative region that was epidemiologically more similar in terms of LBW prevalence (appendix pp 36–37). Penalised splines were used to smooth the trends across the time series at the country level.^{30–32} This smoothing meant that country-level, non-linear time trends were captured in the modelling, without random variation and outliers affecting the estimates. The country-level covariates were also included in the modelling.

The data quality categories were used to apply shifts to the input data to account for biases and additional variance terms. The bias shift was applied to administrative data from lower-quality categories, which approximated the expected bias from heaping that was already accounted for in the survey adjustment. The additional variance was based on the data quality category of the administrative data and the weighting between administrative and survey data if both data sources were available for the country (appendix pp 24–25).

Standard diagnostic checks were used to assess for convergence and the sampling efficiency. Cross-validation was used to validate the model, averaging over 200 random splits of 20% test data and 80% training data. Sensitivity analyses were checks on covariates, bias method, temporal smoothing, and non-informative priors. All models were fitted in R statistical software using the R packages rjags and R2jags (appendix pp 25–30).³³

The model included all 2042 country-years of data that met the inclusion criteria from 158 countries and generated annual estimates of LBW from 2000 to 2020, with 95% credible intervals (CrIs), for 195 countries and areas. We report estimates only for the 158 countries and areas with data. We also produced regional and global aggregates using estimates from all 195 countries and areas (appendix p 32). For the 37 countries and areas with no data or with data that did not meet the inclusion criteria, the final model was used to predict estimates of the LBW prevalence on the basis of country intercepts and

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<https://dhis2.org/about/>

time trends estimated from the region-level and country-level covariates for all country-years. These estimates were used only to produce the regional and global estimates.

The absolute numbers of LBW births were calculated by multiplying the LBW prevalence by the WPP²³ estimate of the number of livebirths. The average annual rate of reduction (AARR) achieved was calculated using the standard formula applied for target tracking by the World Health Assembly:³⁴ $AARR = 100 \times (1 - \exp(\beta))$, where β is the slope of the natural logarithm of the prevalence time series regressed over the time period (ie, 2000–20 or 2012–20; appendix p 32). Because a time series for the target is not available, the target AARRs were calculated using only the baseline and target values in this formula. This calculation yielded a constant reduction of AARR over the target time period—differing from the actual AARRs, which can vary over time owing to the use of the full time series.

Analysis of LBW subgroups

For countries and areas that reported administrative LBW data meeting the inclusion criteria by subgroup (birthweight 2000–2499 g, 1500–1999 g, 1000–1499 g, and <1000 g), we conducted a subgroup analysis to assess the variation in the proportion of LBW contributed by each subgroup by region and over time. The proportion of the total LBW contributed by each subgroup for each country-year was calculated using the formula (number of births in the subgroup)/(total number of LBW newborns). The regional and global proportions were estimated using two methods, as the data had low (<60%) regional coverage due to lack of or non-standardised reporting of the LBW subgroups, especially in southern Asia and sub-Saharan Africa. In method 1, a random effects meta-analysis of single proportions with a logit transformation using the meta-prop function in R was conducted^{35–37} to estimate the subgroup proportions. In method 2, the regional proportion for northern America, Australia and New Zealand, central Asia, and Europe (the only region with >60% coverage) was used. For both methods, the regional and global number of LBW births within each subgroup was calculated by multiplying the proportion by the estimated number of all LBW births (appendix pp 38–46; method 2). Method 1 is presented in the appendix (pp 38–46).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The final dataset comprised 2042 country-years from 158 countries and areas between 2000 and 2020 (with surveys representing births from 1995 to 2020; figure 1, table 1, appendix p 10). Data meeting inclusion criteria

were available for 158 (81%) of 195 countries and areas. Most country-years of data (1706 [83.5%] of 2042) were from national administrative sources, with the majority of these (1546 [90.7%]) coming from high-income (976 [57.3%]) or upper-middle-income (570 [33.5%]) settings. Household surveys were key for ensuring the representation of data from low-income and lower-middle-income countries—all countries in southern Asia and two-thirds of countries in sub-Saharan Africa had at least one adjusted survey estimate included in the input data. No data that met the inclusion criteria were identified for 37 countries and areas.

In 2020, an estimated 19.8 million (95% CrI 18.4–21.7 million) liveborn newborns were LBW globally (table 2). Nearly a quarter (24.4% [20.8–29.3]; 8817000 [7508500–10561700]) of all livebirths in southern Asia in 2020 were LBW; in sub-Saharan Africa, the prevalence was 13.9% (12.4–15.7; 5381300 [4821100–6087700]). These two regions accounted for 55.5% of global livebirths (southern Asia 26.8% [36099000 of 134767000]; sub-Saharan Africa 28.8% [38819300 of 134767000]) but 71.6% of all LBW liveborn newborns (southern Asia 44.5% [8817000 of 19833900]; sub-Saharan Africa 27.1% [5381300 of 19833900]) in 2020. Substantial variation was found between countries: 75 of the 158 countries and areas had an estimated LBW prevalence of less than 10% in 2020, whereas four countries were estimated to have an LBW prevalence of at least 20% (figure 2; appendix pp 47–53).

The estimated global LBW prevalence in 2020 was 14.7% (95% CrI 13.7–16.1; 19833900 [18425700–21694500]), compared with 16.6% (15.5–17.9; 22120400 [20682700–23852400]) in 2000 (table 2). These values show an estimated 11% reduction

	Administrative data (2000–20), number of countries and areas (number of country-years)	National representative survey data (1995–2020), number of countries and areas (number of surveys)	Total data,* number of countries and areas (number of country-years)	Missing data, number of countries and areas
Latin America and the Caribbean (n=33)	21 (306)	20 (70)	27 (376)	6
Eastern Asia, southeastern Asia, and Oceania (n=30)†	12 (149)	15 (45)	22 (194)	8
Northern America, Australia and New Zealand, central Asia, and Europe (n=51)	51 (970)	13 (48)	51 (1018)	0
Southern Asia (n=6)	4 (39)	6 (10)	6 (49)	0
Sub-Saharan Africa (n=48)	13 (73)	32 (127)	35 (200)	13
Northern Africa and western Asia (n=27)	12 (169)	9 (36)	17 (205)	10
Global (n=195)	113 (1706)	95 (336)	158 (2042)	37

SDG regions (revision 1) grouping was adapted for use in the modelling with Iran, Sri Lanka, and Maldives moved from the southern Asia region into the northern Africa and western Asia region. All other countries and areas remain in their original SDG region grouping; the countries and areas included in each regional group are listed in the appendix (pp 36–37). SDG=Sustainable Development Goals. *Some countries have both administrative and survey data. †Excluding Australia and New Zealand.

Table 1: Input data by type and region

	2000		2012		2020		AARR in LBW prevalence, %	
	LBW prevalence, % (95% CrI)	Number of LBW newborns (95% CrI)	LBW prevalence, % (95% CrI)	Number of LBW newborns (95% CrI)	LBW prevalence, % (95% CrI)	Number of LBW newborns (95% CrI)	2000–20	2012–20*
Latin America and the Caribbean	9.3% (9.0–9.6)	1 070 600 (1 037 500–1 109 800)	9.5% (9.2–9.9)	1 017 100 (983 300–1 056 900)	9.6% (9.2–10.1)	944 400 (904 900–992 600)	–0.14%	–0.20%
Eastern Asia, southeastern Asia, and Oceania†	8.7% (8.2–9.4)	2 737 700 (2 574 900–2 934 300)	8.2% (7.8–8.6)	2 728 900 (2 608 900–2 863 200)	8.8% (8.2–9.4)	2 216 200 (2 076 500–2 380 000)	0.33%	–0.59%
Northern America, Australia and New Zealand, central Asia, and Europe	7.3% (7.1–7.5)	960 900 (938 200–988 100)	7.3% (7.2–7.4)	1 051 000 (1 036 700–1 067 100)	7.2% (7.1–7.3)	945 300 (931 500–960 400)	0.04%	0.13%
Southern Asia	28.9% (25.5–32.9)	11 869 000 (10 478 600–13 508 900)	26.1% (22.8–30.4)	10 238 000 (8 941 600–11 916 200)	24.4% (20.8–29.3)	8 817 000 (7 508 500–10 561 700)	0.88%	0.85%
Sub-Saharan Africa	15.7% (14.1–17.7)	4 227 400 (3 796 100–4 769 200)	14.5% (13.1–16.4)	4 998 500 (4 489 100–5 632 100)	13.9% (12.4–15.7)	5 381 300 (4 821 100–6 087 700)	0.63%	0.59%
Northern Africa and western Asia	13.2% (11.0–16.2)	1 254 700 (1 048 400–1 535 400)	13.1% (10.8–16.2)	1 530 000 (1 261 600–1 895 900)	13.1% (10.8–16.4)	1 529 700 (1 254 800–1 909 500)	0.05%	–0.03%
Global	16.6% (15.5–17.9)	22 120 400 (20 682 700–23 852 400)	15.0% (14.0–16.2)	21 563 500 (20 165 400–23 352 800)	14.7% (13.7–16.1)	19 833 900 (18 425 700–21 694 500)	0.75%	0.30%

AARR=average annual reduction rate. CrI=credible interval. LBW=low birthweight. *Average annual rate of reduction from 2012 because this was the baseline year for the target. †Excluding Australia and New Zealand.

Table 2: Estimated LBW prevalence and number of LBW newborns by region for 2000, 2012, and 2020

in the LBW prevalence over this period (AARR 0.75%) and an absolute reduction of 1.9 percentage points (table 2). When measuring from the global target baseline year (2012) to 2020, the actual AARR at the global level was only 0.3%. The estimated AARR between 2012 and 2020 was highest in southern Asia (0.85%) and sub-Saharan Africa (0.59%), whereas Latin America and the Caribbean; eastern Asia, southeastern Asia and Oceania (excluding Australia and New Zealand); and northern Africa and western Asia had a negative AARR (table 2), indicating that the LBW prevalence increased in these regions. Globally, the estimated number of LBW newborns decreased from 22.1 million (95% CrI 20.7–23.9 million) in 2000 to 19.8 million (18.4–21.7 million) in 2020 (table 2, figure 3). The largest decrease in numbers was seen in southern Asia, with 11.9 million (10.5–13.5 million) in 2000 and 8.8 million (7.5–10.6 million) in 2020. Sub-Saharan Africa and northern Africa and western Asia are the only regions in which the number of LBW newborns increased during this period (sub-Saharan Africa 4.2 million [3.8–4.8 million] in 2000 to 5.4 million [4.8–6.1 million] in 2020; northern Africa and western Asia 1.3 million [1.0–1.5 million] in 2000 to 1.5 million [1.3–1.9 million] in 2020).

In the northern America, Australia and New Zealand, central Asia, and Europe region—the only region with data with a regional coverage of more than 60%—approximately 35% of all liveborn LBW newborns weighed less than 2000 g: 5.8% (95% CI 5.2–6.4; 54 800 [49 400–60 800]) weighed less than 1000 g, 9.0% (8.7–9.4; 85 400 [82 000–88 900]) weighed 1000–1499 g, and 19.4% (19.0–19.8; 183 500 [180 000–187 000]) weighed 1500–1999 g. These values correspond to an

estimated 324 000, or 2.5%, of liveborn newborns with a birthweight of less than 2000 g in this region (appendix p 46).

Discussion

We estimated the number of LBW newborns and the LBW prevalence globally in 2020, and time trends from 2000 to 2020, using the largest country dataset available to date, which included ten more countries than were featured in previous estimates.¹⁷ We used a Bayesian modelling approach that was more robust than previous models and included all the data in the same model using the data quality categorisation, which enabled the different data sources to be weighted, biases to be accounted for, and changes in data quality over time to be captured. An estimated 19.8 million LBW newborns were born in 2020, and 449 million LBW newborns were born worldwide between 2000 and 2020.

The Global Nutrition Target called for a 30% reduction in LBW prevalence from the 2012 baseline of 15% to an endline of 10.5% by 2030, requiring an AARR of 1.96% over this time period. Given that the estimated AARR between 2012 and 2020 is only 0.3%, the required AARR between 2020 and 2030 is now 3.31%—an 11-fold increase. Of note is that the AARR was lower in the 2012–20 period than in the 2000–20 period for all regions except northern America, Australia and New Zealand, central Asia, and Europe, suggesting that progress has slowed in the most recent period in regions with the highest LBW prevalence. Although southern Asia (AARR 0.85%) and sub-Saharan Africa (AARR 0.59%) had the largest declines in LBW prevalence between 2012 and 2020, these regions are far from the AARR of 1.96% required to achieve the 2030 target. Furthermore, as

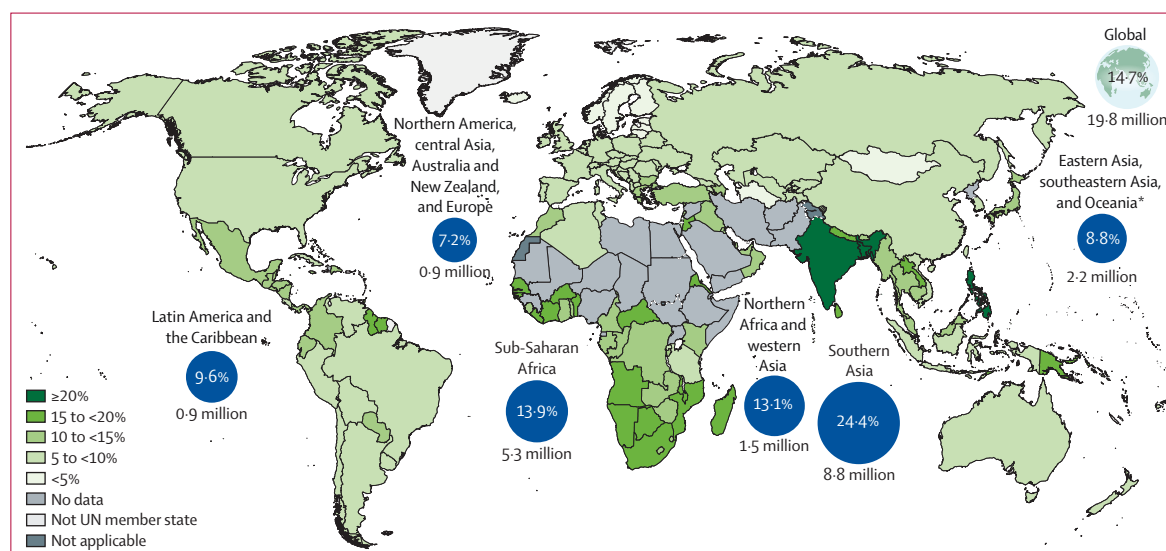


Figure 2: National and regional numbers and LBW prevalence for newborns in 2020

This map is stylised and not to scale and does not reflect a position by UNICEF or WHO on the legal status of any country or territory or the delimitation of any frontiers. Dotted and dashed lines on the map represent approximate border lines for which there might not be full agreement. LBW=low birthweight. *Excluding Australia and New Zealand.

nearly three-quarters of global LBW births occur in these regions (44.5% in southern Asia and 27.1% in sub-Saharan Africa), the global target will not be met without concerted efforts to prevent LBW focused on the two underlying pathways: preterm birth and intrauterine growth restriction.^{38,39} In addition to implementing programmes for vulnerable girls and women with undernutrition and anaemia, other important risk factors to address include infections and obstetric causes.^{38,40}

Improving on our previous estimates, here we were able to report birthweight distributions by subgroup for the first time. We estimated the distribution for LBW subgroups for the northern America, Australia and New Zealand, central Asia, and Europe region, noting that two-thirds (65.6% [95% CI 64.7–66.5]) of LBW births were in the 2000–2499 g category, with 19.4% (19.0–19.8) in the 1500–1999 g category, 9.0% (8.7–9.4) in the 1000–1499 g category, and only 5.8% (5.2–6.4) in the less than 1000 g category; however, those newborns weighing less than 1000 g are the most vulnerable and most are of lower gestational age (figure 4). This LBW subgroup information is helpful for individual risk stratification on important outcomes such as mortality risk, and at a population level for planning appropriate intervention programmes. We note that the subgroup distributions showed little variation and are stable between regions and over time, although these data mainly came from high-income countries (appendix p 36).

Although routine data systems are improving in many countries, gaps remain—especially in regions with high burden of LBW. Compared with previous estimates, this study includes LBW data for ten additional countries, now covering more than 80% of the WHO Member

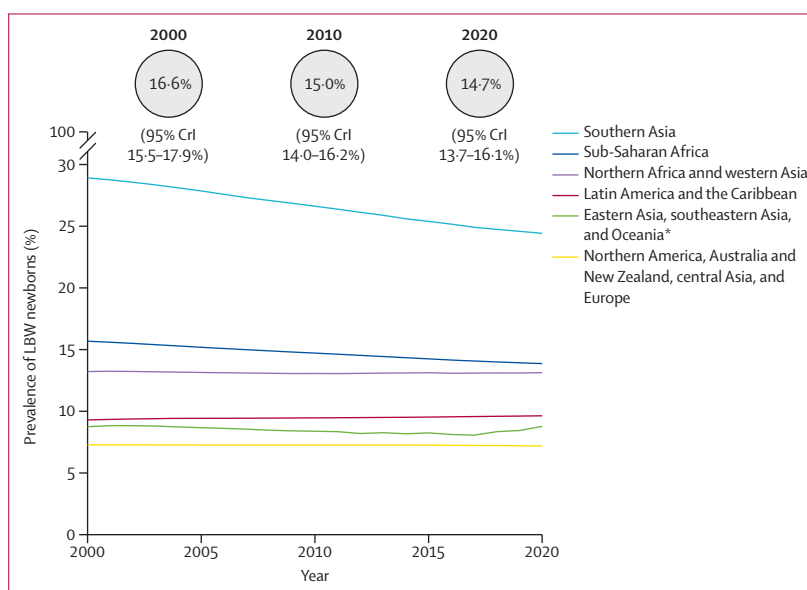


Figure 3: Regional and worldwide trends in LBW prevalence between 2000 and 2020

Worldwide estimates (with 95% CrIs) for 2000, 2010, and 2020 are shown in the circles. See appendix pp 37–38. CrI=credible interval. LBW=low birthweight. *Excluding Australia and New Zealand.

States and the occupied Palestinian territory, including east Jerusalem (158 of 195 countries and areas). The highest capture of LBW data was in the region of northern America, Australia and New Zealand, central Asia, and Europe, where nearly 85% of livebirths (according to WPP 2022 estimates) were accounted for in the most recent administrative data between 2000 and 2020. In sub-Saharan Africa, the number of countries with administrative LBW data meeting

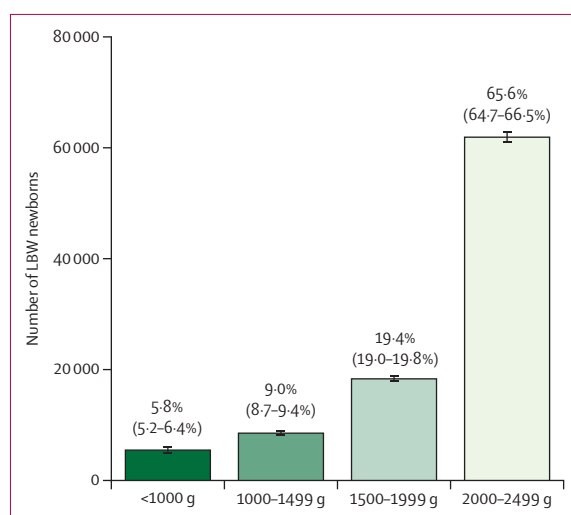


Figure 4: Number of LBW newborns by subgroup for northern America, Australia and New Zealand, central Asia, and Europe in 2020 (n=945 300)
Error bars represent 95% CIs. Data above the bars show the percentages (with 95% CI) of LBW newborns in each subgroup. See appendix pp 38–46. LBW=low birthweight.

inclusion criteria more than tripled from just four countries in the previous 2015 estimates to 13 in the current estimates. However, with increasing facility births and with more countries and areas starting to apply information technologies in their health systems, substantial progress in availability of LBW data can be achieved relatively quickly, although concerted efforts will be needed to ensure that any increases in data availability are linked with efforts to improve quality.

Strengths in these estimates include the larger dataset, with 593 more country-years of data than in the previous estimates; the number of countries and areas with included survey data has also increased from 86 to 95. In these current estimates, a Bayesian framework allowed both types of source data to be incorporated in one model, accounting for regional differences and data quality biases and enabling all the data to be used in estimating LBW, thereby strengthening the country-level, regional, and global estimates and resulting in comparable LBW trends between countries. By comparison, two separate models were used in the 2015 estimates depending on data quality, so the data available were not all used to their full advantage.

Despite these advances, important limitations remain. First, for 37 countries and areas, no data meeting the inclusion criteria were available—all were low-income or lower-middle-income countries and many were experiencing humanitarian crises. Second, the administrative input data also have limitations including a lack of access to individual-level data, meaning that we were unable to consistently assess and adjust for potential heaping and missingness. Our data quality categorisation attempted to account for this lack of information by grouping countries and areas according to data quality indicators; however,

more robust methods need to be developed to adjust for quality differences in administrative data at an individual-country level, as opposed to having a single bias adjustment for a group of countries. For surveys, the standard errors are larger than those developed for the administrative input data owing to the nature of sampling in household surveys. These differences in standard errors between administrative and survey data could affect the model outcome artificially. Third, although the SDG geographical groupings used in the modelling are more similar epidemiologically than other regional groupings and we sought to adjust the groups for known regional outliers (eg, Iran, Sri Lanka, and Maldives), other regional outliers could remain, for which the estimated LBW might be less reliable.

Finally, an additional limitation is the absence of subgroup data for the majority of births in the world. The subgroup dataset we used covered only 13% of global births: 0% of births in low-income countries, 1.7% of births in lower-middle-income countries, 24.7% of births in upper-middle-income countries, and 78.4% of births in high-income countries. From a regional perspective, subgroup data were available for less than 0.2% of all births in sub-Saharan Africa and 0% of births in southern Asia. Therefore, the estimates for most regions might not be accurate.

Countries need reliable and timely data to target interventions and monitor progress towards the Global Nutrition Target.¹⁰ We observed encouraging progress in data availability made by the sub-Saharan Africa and southern Asia regions over the past decade. However, we also noted that major gaps still exist between regions (table 1). Advocating for capturing information about LBW subgroups is also important, given that data availability was a limitation in most of the regions. In settings in which routine information systems that capture and report on birthweight data remain unavailable or of low quality, household surveys continue to be a crucial source of LBW data.⁴¹ With more than 80% of global births now in facilities, and with the potential for birthweight data to be included among the data routinely captured by facilities,¹⁵ investing in strengthening the routine health information systems is crucial.^{42,43} National facility-based information platforms such as DHIS2 as well as CRVSs can all capture data on birthweight, but require strengthening in many countries and areas. For example, digital weighing devices are known to be more accurate than analogue devices and their use can result in less birthweight heaping, but appropriate national standards and the training and supervision of health-care providers tasked to measure and record birthweights are equally important. Efforts to strengthen national standards and available weighing equipment will be key actions if accuracy and coverage of birth weighing and birthweight reporting are to be improved.

Furthermore, consistent with previous global estimates and global nutrition monitoring targets, the LBW

estimate is for livebirth only and stillbirth was not included. However, in the future, we suggest capturing the whole burden associated with fetal vulnerability, including stillbirths as well as liveborn neonates with preterm or LBW.⁴⁴

Insufficient progress has been made in LBW reduction in the past two decades and, if current trends continue, the Global Nutrition Target of a 30% reduction in LBW prevalence between 2012 and 2030 will not be achieved. To meet this target, considerable efforts are needed to yield an 11-fold increase in AARR from the observed 0.3% to the required 3.31%. With increasing facility births and technological advances, improvements in data quality and availability are achievable. Countries need to prioritise programme investments to prevent LBW throughout the lifecycle, include stillbirths as an outcome,⁴⁴ and increase focus on preterm birth⁴⁵ and intrauterine growth restriction as the underlying pathways of LBW, in addition to improving data quality and data use for accountability.³⁸

Contributors

EBo, CH, and JEL, with EOO and HB, had the idea and led the overall technical coordination of the process. YBO coordinated the initial administrative database, covariates database, and data quality assessment, with contributions from HB, JK, JC, BK, and GG-D. JK coordinated the survey analysis, with contribution from JC and CC. EBr led the Bayesian modelling with overall statistical oversight from EOO, with contributions from HB, JK, GAS, AL, JC, CC, JEL, EBo, and CH. DGEF, JK, EOO, EBr, CH, A-BM, GG-D, and LH-A contributed to country consultation coordination. All authors reviewed and helped to revise the manuscript and agreed with the final version. The authors alone are responsible for the views expressed in this Article and they do not necessarily represent the views, decisions, or policies of the institutions with which they are affiliated. All authors had full access to all data in the study and had final responsibility for the decision to submit for publication. YBO, JK, and EBr accessed and verified the data.

Declaration of interests

We declare no competing interests.

Data sharing

Data sharing and transfer agreements were jointly developed and signed by all collaborating partners. Pooled summary tables generated have been deposited online at <https://github.com/WorldHealthOrganization/mca-pretermbirths-analysis>, with data access subject to approval by collaborating parties.

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